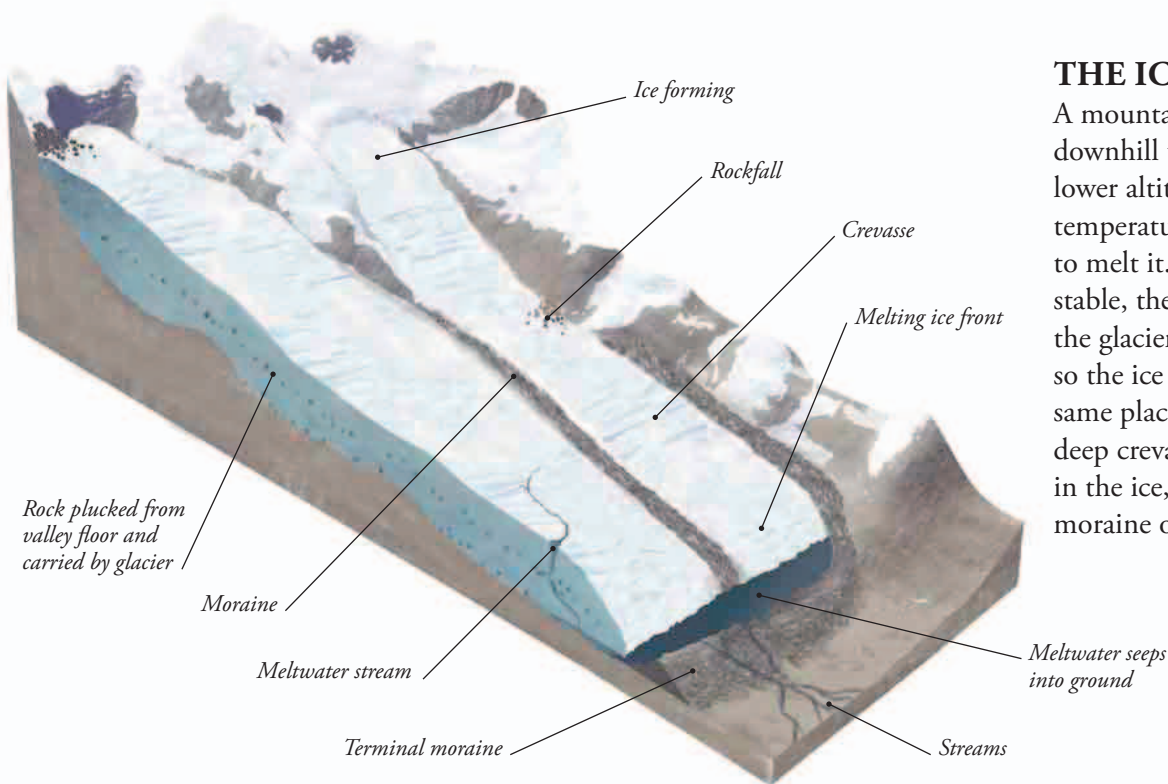




## THE ICE FRONT

A mountain glacier flows downhill until it reaches a lower altitude where the air temperature is warm enough to melt it. If the climate is stable, the ice melts as fast as the glacier moves forward, so the ice front stays in the same place. It is marked by deep crevasses (large cracks) in the ice, and a terminal (end) moraine of rock debris.

## ICE SHEETS

Antarctica and Greenland are covered by incredibly deep ice sheets. The East Antarctic Ice Sheet forms a huge dome up to 2.8 miles (4.5 km) thick, and its weight has made the rock beneath it sink as much as 0.6 miles (1 km) into Earth's crust. Parts of the Antarctic ice sheets extend over the sea as floating ice shelves.

### ICE SHELF ►

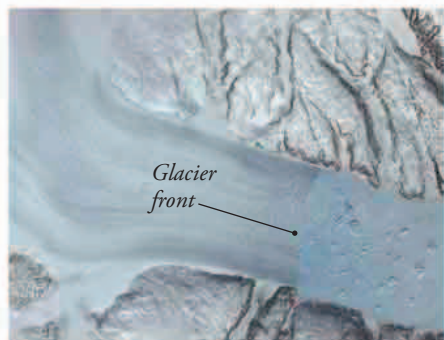
*The sheer cliffs of the Antarctic ice shelves are up to 160 ft (50 m) high.*



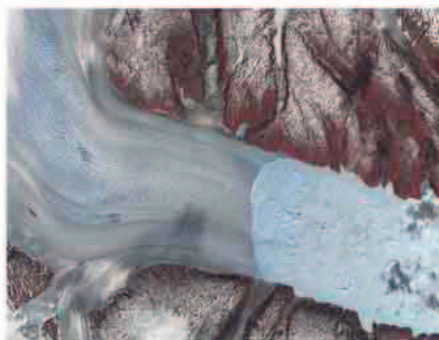
## RETREATING GLACIERS

Global warming is making most of the world's glaciers melt away. The lower stretches of many mountain glaciers have disappeared, so they seem to have retreated uphill. These satellite images show how the front of the Helheim Glacier in Greenland retreated 4.2 miles

(7.5 km) between 2001 and 2005 (in each image, the glacier is on the left). Some coastal glaciers are breaking up, including several Antarctic ice shelves, and all the extra ice tumbling into the ocean is raising sea levels.



2001



2003



2005